

PRATHAP SELVAKUMAR

Manchester, UK | +44 7586 438575 | prathapsk@hotmail.com | linkedin.com/in/prathapsk | prathapsk.vercel.app | github.com/prathapselvakumar

PERSONAL STATEMENT

MSc Robotics student at the University of Manchester with a strong foundation in AI, computer vision, and autonomous systems. Experienced in designing and deploying intelligent robotics pipelines using Python, ROS2, and deep learning frameworks. Published researcher with industry experience across software engineering and AI/ML. UK Graduate Visa holder — no sponsorship required.

WORK EXPERIENCE

C-Square Info Solutions (Reliance Retail Subsidiary)

Aug 2024 – May 2025

Software Engineer · Remote

- Re-engineered 50+ static APIs into dynamic, modular endpoints, eliminating over 1,000 redundant lines of code per report page and improving system scalability by 40%.
- Redesigned front-end architecture using reusable React hooks and context APIs, increasing interface responsiveness and maintainability by 25%.
- Optimised database queries and implemented secure RESTful API authentication, reducing data retrieval time by 30%.

Qentelli Solutions

Feb 2024 – Apr 2024

AI/ML Intern · Hyderabad

- Designed and deployed TensorFlow-based automation models, reducing manual task effort by 40%.
- Processed and optimised over 100,000 training data entries, improving pipeline speed and data quality by 35%.
- Evaluated multiple classification models, achieving 92% accuracy on predictive tasks

National University of Singapore

Dec 2023 – Jan 2024

Global Academic Intern — Big Data Analytics using Deep Learning · Singapore

- Led a team of 4 interns to deliver a deep learning project, achieving a final score of 44/50 (Grade A).
- Trained and fine-tuned neural networks on large-scale datasets, improving model precision by 15%.
- Built Python-based data visualisation dashboards (Matplotlib, Seaborn), reducing analysis time by 25%.

PROJECTS

Autonomous Rover Development (Leo Rover)

Oct 2025 – May 2026

- Designed and deployed a ROS2-based robotic system achieving 95% autonomous navigation accuracy in indoor and semi-outdoor environments.
- Collaborated in a 5-member team to calibrate control algorithms, reducing positional drift by 20%.
- Built a RealSense D435i computer vision pipeline for coloured block detection using HSV masking, computing 3D drop coordinates for robotic arm manipulation.

Drone Feedback Controller

Mar 2026 – Apr 2026

- Designed a hybrid Q-learning and nested PID control system for UAV stabilisation, with an online RL agent selecting optimal gain profiles in real time from discretised flight states.

- Implemented experience replay using a 400-sample circular buffer for stable off-policy TD (0) updates, with adaptive integral clamping tuned for wind disturbance rejection.

A* Path Planning System

Mar 2026 – Apr 2026

- Implemented an interactive A* path planning system with dynamic obstacle placement, route replay, and heuristic-based shortest-path search.
- Built a React/TypeScript interface with live grid resizing, path validation, and cost diagnostics.

Agro Analytics — Agricultural AI Pipeline

Dec 2023 – Jan 2024

- Built an AI-driven crop monitoring pipeline using satellite, soil, and sensor data for predictive agricultural analysis.
- Developed Random Forest models that improved crop yield forecasting accuracy by 12% over baseline methods.

Snake Detection — Computer Vision

Jul 2022 – Aug 2022

- Trained a custom YOLOv8 model for real-time snake detection and venomous/non-venomous classification with 94% confidence.
- Integrated GPS-based alert logging to support rapid location-aware safety notifications.

Audio Search Engine — Published Research

Apr 2022 – May 2022

- Developed an audio search engine using unsupervised clustering and feature extraction for similarity-based retrieval across large datasets.
- Published peer-reviewed research in IGI Global on scalable audio indexing: "Comparative Analysis Implementation of Queuing Songs in Players Using Audio Clustering Algorithm".

TECHNICAL SKILLS

- **Robotics & Simulation:** ROS2, PyBullet, Gazebo, MoveIt2, Nav2, Intel RealSense D435i
- **AI / Machine Learning:** TensorFlow, PyTorch, Stable-Baselines3, YOLOv8, Scikit-learn, OpenCV
- **Programming:** Python, C++, MATLAB, JavaScript (React, Node.js), TypeScript
- **Cloud & Data:** AWS SageMaker, Pandas, Matplotlib, Seaborn, Excel (advanced, pivot tables)
- **Other:** Git, RESTful APIs, Agile/Scrum, sensor integration, algorithm design, image processing
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EDUCATION

University of Manchester

2025 – 2026 (Expected)

MSc Robotics

- Relevant modules: Control Systems, Multi-Agent Reinforcement Learning, Autonomous Systems, Digital Manufacturing, Computer Vision.
- Used MATLAB, TensorFlow, and OpenCV to simulate and automate robotic tasks, reducing testing time by 30%.

SRM Institute of Science and Technology

2020 – 2024

BEng Computer Science and Engineering — First Class with Distinction, CGPA: 8.49 / 10

- Applied AI, machine learning, and data science methodologies across six applied research and development projects.
- Built an AI-based image recognition system for automation, increasing classification accuracy by 12% and reducing manual verification time by 25%.

PUBLICATIONS

Selvakumar, P. (2022). *Comparative Analysis Implementation of Queuing Songs in Players Using Audio Clustering Algorithm*. IGI Global. Peer-reviewed journal article.